AI Synthesis Documentation

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AI Use 1

Before

I already had the main idea of the project and understood that I wanted to analyze emotional tone in Reddit relationship_advice posts using valence and arousal. However, I wanted to clarify how to explain the VAD method in simple academic language. I needed to make sure that the difference between valence and arousal was clear and that the method was described correctly in the paper.

The Prompt

Can you help me explain the difference between valence and arousal for a student-level Python project? I am using the NRC VAD Lexicon to analyze emotional tone in Reddit posts.

After

The AI assistant helped me clarify the explanation of the two emotional dimensions. I used the explanation that valence shows whether a text is more positive or negative, while arousal shows emotional intensity. This helped me describe the method more clearly in the Background and Methodology sections of the paper. The project idea stayed the same, but the explanation became easier to understand.

AI Use 2

Before

During the project, I noticed that the dataset needed to be more consistent for comparing time-of-day groups. I wanted to make sure that each group, morning, afternoon, evening, and night, had the same number of posts so that the comparison would be fair.

The Prompt

How can I make my dataset more balanced for comparing Reddit posts across morning, afternoon, evening, and night?

After

The AI assistant suggested using equal sample sizes for each time-of-day group. I used this idea to prepare a dataset with 500 posts for each group: morning, afternoon, evening, and night. This made the comparison between groups clearer and more consistent.

AI Use 3

Before

There were problems with reading and organizing the CSV file. Some column names were not always recognized correctly, so the script needed a more flexible way to identify the text and timestamp columns.

Before code example:

```
df = pd.read_csv(input_csv)

text = df["selftext"]
created_time = df["created_utc"]
```

This version was too strict because it expected the CSV file to always have the exact same column names.

The Prompt

My Python script has problems reading the CSV file because some columns are not recognized consistently. How can I make the CSV loading and column handling more flexible?

After

The AI assistant suggested checking column names more carefully and using helper functions to find the correct columns. I used this idea to improve the CSV loading part of the code.

After code example:

```
def clean_column_name(name):
name = str(name).lower().strip()
name = name.replace(";", "")
name = re.sub(r"[^a-z0-9_]", "", name)
return name

def find_col(df, names):
mapping = {clean_column_name(c): c for c in df.columns}

for name in names:
clean_name = clean_column_name(name)
if clean_name in mapping:
return mapping[clean_name]

return None

text_col = find_col(
df,
["text", "body", "comment", "content", "selftext", "post_text",
"description"]
)

time_col = find_col(
df,
["created_utc", "timestamp", "date", "datetime", "created_at", "time",
"created"]
)
```

This made the script more reliable because it could handle small differences in column names. After this change, the dataset could be loaded and prepared more consistently.

AI Use 4

Before

The timestamp classification was important because the project depended on grouping Reddit posts into morning, afternoon, evening, and night. At first, the code treated timestamps too simply, which could cause incorrect hour values if the timestamp format was different.

Before code example:

```
df["created_time"] = pd.to_datetime(df["created_utc"], unit="s")
df["hour_utc"] = df["created_time"].dt.hour
```

This worked only if the timestamp was already in seconds. It was not flexible enough for other numeric timestamp formats.

The Prompt

Can you help me check the timestamp logic in my Python project? I need to convert Reddit timestamps correctly and classify posts into morning, afternoon, evening, and night.

After

The AI assistant helped me review the timestamp conversion logic. I used the suggestion to make the code check the size of the numeric timestamp before converting it.

After code example:

```
raw = df[time_col].astype(str).str.replace(";", "", regex=False).str.strip()
numeric = pd.to_numeric(raw, errors="coerce")

if numeric.notna().mean() > 0.7:
median_value = numeric.dropna().median()

if median_value > 10 ** 17:
created_utc = numeric / 10 ** 9
elif median_value > 10 ** 14:
created_utc = numeric / 10 ** 6
elif median_value > 10 ** 12:
created_utc = numeric / 1000
else:
created_utc = numeric

return created_utc.astype("float")
```

This made timestamp handling more reliable. The script could process timestamps in different numeric formats before assigning posts to time-of-day groups.

AI Use 5

Before

After running the code, I had the final tables, figures, and statistical results. However, I wanted to make sure that the Results and Discussion sections were separated properly. I was not fully sure which information should stay in Results and which information should be explained in Discussion.

The Prompt

Can you help me understand what should go in the Results section and what should go in the Discussion section for my Python final project? I have VAD scores, statistical tests, figures, hypotheses, and limitations.

After

The AI assistant helped me separate the two sections more clearly. I used the Results section to describe the actual findings: mean valence, mean arousal, p-values, and what the tables and figures showed. I used the Discussion section to explain what the findings meant, whether the hypotheses were supported, why the results may have appeared this way, and what the project limitations were. This helped make the paper more organized and easier to follow.